

(20 Points) Consider a One-Time Pad over $\mathbb{Z}_5 = \{0, 1, \dots, 4\}$ where $C \equiv M + K \pmod{5}$. The key distribution is: $\Pr[K = 0] = 0.1$, $\Pr[K = 1] = 0.2$, $\Pr[K = 2] = 0.3$, $\Pr[K = 3] = 0.4$ (0 for others). The message distribution is $\Pr[M = 0] = 1/6$, $\Pr[M = 1] = 1/3$, and $\Pr[M = 2] = 1/2$.

- (a) Does this scheme satisfy Perfect Secrecy for the key distribution above?
- (b) Calculate $\Pr[M = 0 \mid C = 1]$.
- (c) Calculate $\Pr[M = 2 \mid C = 3]$.
- (d) Calculate $\Pr[M = 1 \mid C = 0]$.

(5 Points) Consider a symmetric-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ over the message space $\mathcal{M} = \{0, 1\}^n$. The scheme uses a secure PRF $F : \{0, 1\}^n \times \{0, 1\}^5 \rightarrow \{0, 1\}^n$.

The encryption algorithm $\text{Enc}_k(m)$ is defined as:

$$c = (r, m \oplus F(k, r))$$

where r is sampled uniformly from $\{0, 1\}^5$.

Is this encryption scheme CPA-secure?

- A. Yes.** The proof is by reduction. If there exists a PPT adversary $\mathcal{A}$ that distinguishes c from a truly random string, we can construct a PPT distinguisher $\mathcal{D}$ that breaks the pseudorandomness of F by embedding its own challenge r into the encryption oracle, simulating $c = (r, F(k, r) \oplus m)$ for $\mathcal{A}$.
- B. Yes.** The scheme is CPA-secure because F is a secure PRF, and the XOR operation ensures that the ciphertext is computationally indistinguishable from random strings for any polynomial number of queries.
- C. No.** The scheme is insecure because an adversary can use an algebraic attack to break the PRF and recover the secret key k with non-negligible advantage.
- D. No.** Because in its CPA game queries, there is a non-negligible probability that the random value r used in the challenge ciphertext c has already appeared in a previous encryption query.

(5 Points) Let $F : \{0, 1\}^n \times \{0, 1\}^m \rightarrow \{0, 1\}^m$ be a secure PRF. Alice constructs a new function $F' : \{0, 1\}^n \times \{0, 1\}^m \rightarrow \{0, 1\}^m$ defined as:

$$F'(k, x) = F(k, 1^m \oplus x) \oplus x$$

where $k \in \{0, 1\}^n$ is the secret key and $x \in \{0, 1\}^m$ is the input.

Is F' a secure PRF?

- **A. Yes.** The proof is by reduction. If there exists a PPT adversary $\mathcal{A}$ that distinguishes F' from a random function, we can construct a PPT distinguisher $\mathcal{D}$ that breaks the pseudorandomness of F by simulating the $\oplus x$ operation on F 's oracle outputs.
- **B. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{D}$ can break a single instance of F , we can trivially extend this attack by xor it to successfully distinguish the entire output of F' .
- **C. No.** The construction is insecure because x is known to the adversary. By observing $y = F'(k, x)$, an adversary can recover the internal PRF value as $F(k, x) = y \oplus x$, which allows for a successful algebraic attack.
- **D. No.** The construction violates the definition of a PRF because a PRF must be a deterministic function of the key and input.

(20 Points) Consider a One-Time Pad over $\mathbb{Z}_{1048576}$ where $C \equiv M+K \pmod{1048576}$. The key K is uniform on $\mathbb{Z}_{1048576}$. The message distribution is $\Pr[M = 1024] = 0.37$, $\Pr[M = 2048] = 0.41$, and $\Pr[M = 4096] = 0.22$.

- (a) Does this scheme satisfy Perfect Secrecy for the key distribution above?
- (b) Let $a/b := \Pr[C = 5000 \mid M = 1024]$. Calculate (a, b) . (the numerator and denominator must be reduced to their simplest form)
- (c) Calculate $\Pr[M = 2048 \mid C = 2048]$.
- (d) Calculate $\Pr[M = 4096 \mid C = 1]$.

(10 Points) Consider the encryption scheme $\Sigma^n = (\text{Gen}, \text{Enc}, \text{Dec})$ defined over $\mathbb{Z}_n$, where $n \in \{2^k \mid k \in \mathbb{Z}^+\}$ (i.e., 2, 4, 8, ...):

- $\mathcal{M} = \mathcal{K} = \mathcal{C} = \mathbb{Z}_n \times \mathbb{Z}_n$.
- Gen : Sample a uniform pair $(k_1, k_2) \leftarrow \mathbb{Z}_n^2$.
- Enc : For input $(m_1, m_2) \in \mathbb{Z}_n \times \mathbb{Z}_n$, output $c = (k_1 + 3m_1 \pmod{n}, k_2 + 3m_2 \pmod{n})$.

- (a) Set $n := 16$, and let $a/b := \Pr[M = (3, 11) \mid C = (5, 7)]$. Calculate (a, b) . (the numerator and denominator must be reduced to their simplest form)
- (b) Determine whether the scheme described above is a valid secure encryption scheme.

- A. Yes.** First, the key is chosen uniformly at random from $\mathcal{K}$. Second, for this scheme, the probability $\Pr[C = c \mid M = m]$ is exactly $1/|\mathcal{K}|$, which is independent of the message m .
- B. Yes.** The scheme is perfectly secure because the size of the key space, the message space and the ciphertext space are all the same, which is the basic requirement of perfect secrecy in Shannon's theorem.
- C. No.** Due to its correctness does not hold for the entire space.
- D. No.** The encryption function is linear with respect to the message and the key, which allows for algebraic attacks (such as solving systems of linear equations) that can recover the key.

(5 Points) Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{n+1}$ be a secure PRG. Bob constructed a PRG $G' : \{0, 1\}^n \rightarrow \{0, 1\}^\ell$ as follows:

- Sample $s_0 \leftarrow \{0, 1\}^n$.
- For $j = 1, \dots, \ell$: compute $s_j b_j \leftarrow G(s_{j-1})$, where b_j is the last bit.
- Output the string $y = b_1 b_2 \dots b_\ell$.

Is G' a secure PRG?

- **A. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{A}$ distinguishes G' from random, we can build a distinguisher $\mathcal{B}$ for G that embeds its own challenge z to break G via hybrid argument.
- **B. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{D}$ can break a single instance of G , we can trivially extend this attack by run several G to successfully distinguish the entire output of G' .
- **C. No.** The construction violates the definition of G due to uses it incorrectly with mismatch length.
- **D. No.** The construction allows for a distinguisher with non-negligible advantage in polynomial time.

(5 Points) Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{3n}$ be a secure PRG. Let $\text{Trunc}_k(\cdot)$ denote the function that outputs the first $2n$ bits of its input. Carol constructed a $G^* : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ defined as follows: On input seed $s \in \{0, 1\}^n$, output:

$$G^*(s) := \text{Trunc}_{2n}(G(s)).$$

Is G^* a secure PRG?

- **A. Yes.** The proof is by reduction: assume there exists a PPT adversary $\mathcal{A}$ that distinguishes $G^*(s)$ from a uniform string $z \in \{0, 1\}^{2n}$, so it can be used by a distinguisher $\mathcal{B}$ to break the underlying PRG G . $\mathcal{B}$ receives a challenge $w \in \{0, 1\}^{3n}$, passes $\text{Trunc}_{2n}(w)$ to $\mathcal{A}$, and outputs $\mathcal{A}$'s guess.
- **B. Yes.** The proof is by reduction: assume there exists a PPT adversary $\mathcal{B}$ that distinguishes $G(s)$ from a uniform string $z \in \{0, 1\}^{3n}$, so it can be used by a distinguisher $\mathcal{A}$ to break the PRG G^* . $\mathcal{A}$ receives a challenge $w \in \{0, 1\}^{2n}$, pads the challenge with 0 to be length $3n$, passes the result to $\mathcal{B}$, and outputs $\mathcal{B}$'s guess.
- **C. No.** The construction is insecure because truncation is a lossy operation. By discarding n bits of the original PRG output, the entropy of the resulting string $G^*(s)$ is reduced, making it distinguishable from a truly random $2n$ -bit string.
- **D. No.** The construction violates the PRG definition because G^* must use the entire output of G to maintain the computational indistinguishability property inherited from the security parameter λ .

(5 Points) Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ be a secure PRG. Alice constructed a $G^{**} : \{0, 1\}^n \rightarrow \{0, 1\}^{16n}$, the output is defined as:

$$G^{**}(x) := G(G(G(G(X)))) \oplus 1^{16n}.$$

Is G^{**} a secure PRG?

- **A. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{A}$ distinguishes G^{**} from random, we can build a distinguisher $\mathcal{B}$ for G that embeds its own challenge z to break G .
- **B. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{D}$ can break a single instance of G , we can trivially extend this attack by run several G to successfully distinguish the entire output of G^{**} .
- **C. No.** The construction allows for a distinguisher with non-negligible advantage in polynomial time.
- **D. No.** The construction violates the definition of G due to uses it incorrectly with mismatch length.

(5 Points) Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{4n}$ be a secure PRG with output denoted as $k_1 || k_2 || k_3 || k_4$, where $k_1, k_2, k_3, k_4 \in \{0, 1\}^n$. Bob constructed a PRG $G' : \{0, 1\}^n \rightarrow \{0, 1\}^{4n}$ as follows:

- Obtain $k_1 || k_2 || k_3 || k_4 := G(s)$.
- Output $(k_1 \oplus k_2 \oplus k_3) || (k_2 \oplus k_3 \oplus k_4) || (k_3 \oplus k_4 \oplus k_1)$.

Is G' a secure PRG?

- **A. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{A}$ distinguishes G' from random, we can build a distinguisher $\mathcal{B}$ for G that embeds its own challenge z to break G via hybrid argument.
- **B. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{D}$ can break a single instance of G , we can trivially extend this attack by run several G to successfully distinguish the entire output of G' .
- **C. No.** The construction violates the definition of G due to uses it incorrectly with mismatch length.
- **D. No.** The construction allows for a distinguisher with non-negligible advantage in polynomial time.

(5 Points) Let $G : \{0, 1\}^\lambda \rightarrow \{0, 1\}^{2\lambda}$ be a length-doubling secure PRG. We parse the output as $G(s) = G_0(s) \parallel G_1(s)$. We construct a PRF $F : \{0, 1\}^\lambda \times \{0, 1\}^n \rightarrow \{0, 1\}^\lambda$ (where the input length is n). The construction works by traversing a binary tree of depth n defined by the input bits $x = x_1 \dots x_n$.

$$F(k, x) = G_{x_n}(\dots G_{x_2}(G_{x_1}(k)) \dots)$$

Is F a secure PRF?

- **A. Yes.** The proof is by a hybrid argument over the depth n of the tree. We define $n + 1$ hybrid distributions where, in hybrid i , all nodes at depth i are replaced with truly random strings. The indistinguishability between hybrid i and $i + 1$ reduces to the security of the underlying PRG G .
- **B. Yes.** The construction is secure because each level of the tree traversal utilizes a secure PRG. Since a secure PRG is inherently CPA-secure against any polynomial-time adversary, the sequential composition of these evaluations preserves the IND-CPA property for the resulting function F .
- **C. No.** While a PRF requires a randomly sampled secret key k , a PRG must be a strictly deterministic algorithm; therefore, the PRG $G_{x_1}(k)$ cannot take the randomly chosen key k as an input.
- **D. No.** Use the same PRG in a construction is insecure, due to there exists some fixed point which will appear after multiple use of $G(\cdot)$.

(20 Points) Let $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a secure PRF. We define a new function $F' : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ as:

$$F'(k, x) = F(k, x) \oplus F(k, x \oplus 1^n).$$

The PRF Distinguishing Game:

- **World 0** ($b = 0$): The challenger samples a uniform key $k \leftarrow \{0, 1\}^n$ and computes $y_i = F'(k, x_i)$ for each query ¹.
- **World 1** ($b = 1$): The challenger samples a truly random function $f \leftarrow \text{Funs}[\{0, 1\}^n, \{0, 1\}^n]$ and computes $y_i = f(x_i)$ for each query.

Alice proposes an adversary $\mathcal{A}_{\text{Alice}}$ to interact with the challenger and guess b . For $n = 3$, her strategy is as follows:

- (a) Alice chooses a random $x_0 \in \{0, 1\}^3$.
- (b) She constructs 2 additional distinct values x_1, x_2 , where $x_i = x_0 \oplus (0^{3-i}1^i)$. (Note: $0^{3-i}1^i$ denotes $3 - i$ zeros followed by i ones. e.g., $x_1 = x_0 \oplus 001$).
- (c) She queries the challenger for the set $\{x_0, x_1, x_2\}$ and receives $\{y_0, y_1, y_2\}$.
- (d) If she finds any i, j such that $y_i = y_j$ (for $i \neq j$), she outputs $b' = 0$; otherwise, she outputs $b' = 1$.

Questions:

- (a) Calculate $P_1 = \Pr[b' = 0 \mid b = 1]$, the probability that Alice outputs 0 in World 1. (Note: any negligible function can be ignored in your answer, your answer should be only numeric, similar to that in your homework)
- (b) Calculate $P_0 = \Pr[b' = 0 \mid b = 0]$, the probability that Alice outputs 0 in World 0. (Note: any negligible function can be ignored in your answer, your answer should be only numeric, similar to that in your homework)
- (c) Calculate Alice's advantage $Adv = |P_1 - P_0|$.
- (d) Do you think F' is a secure PRF?